

ScholarMotion

A Personalised Visual Tutor for Every
Classroom, Built on Exasol

Exasol AI Build Challenge 2026
Track 4 — AI for the Real World

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The problem, in one number

1,10,971

schools in India are run by a single teacher

40 lakh

children study in them
right now

15.17%

teacher vacancy rate

10 lakh

posts vacant; about 7.5
lakh at primary level

That teacher is not failing. They are outnumbered. When a child does not understand something the first time, there is simply no second explanation available.

Why Track 4 — AI for the Real World

THE REACH ALREADY EXISTS

40.8 crore

rural internet subscribers
(March 2025)

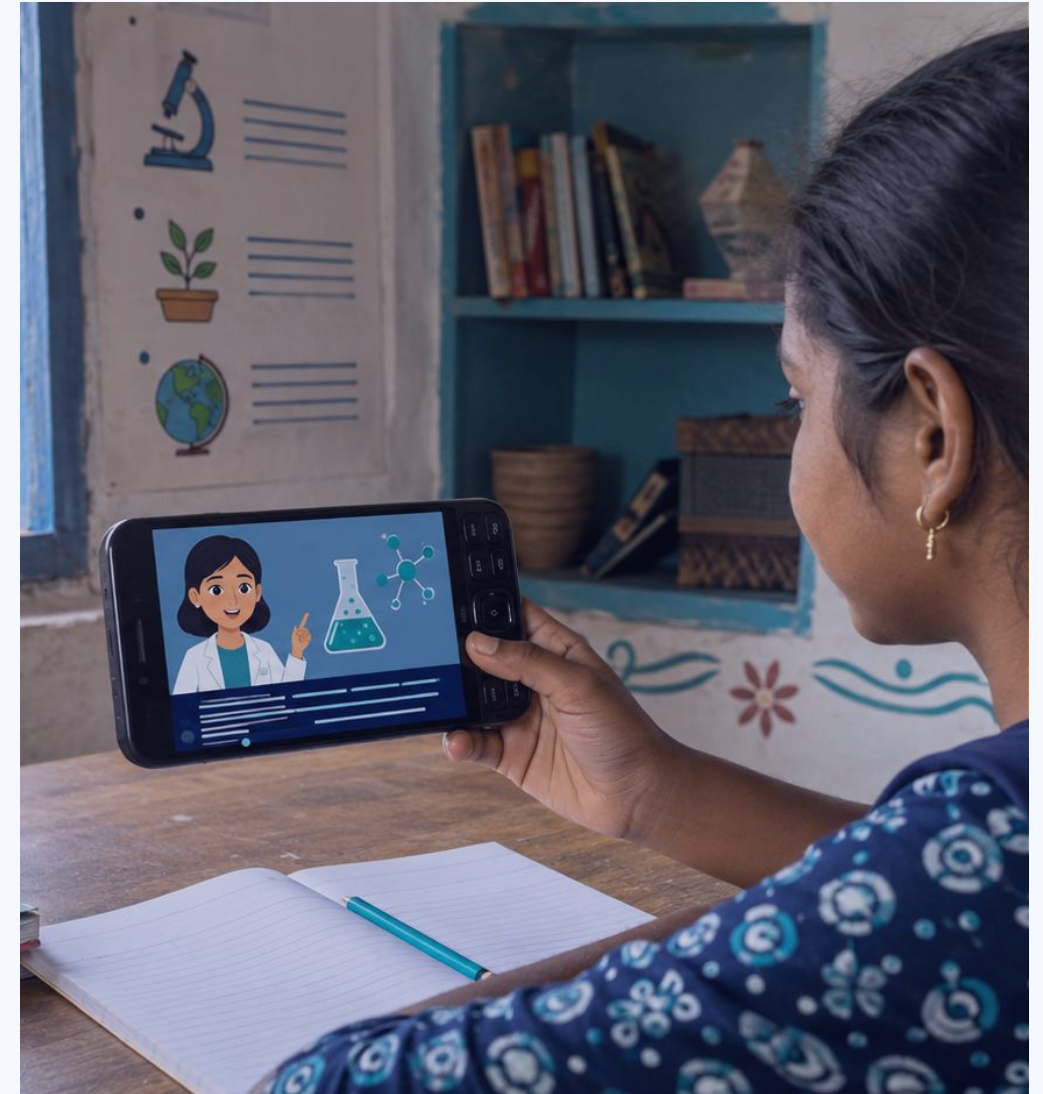
45%

rural internet density —
growing, but still the gap

**What is missing at the far end of that connection is not
bandwidth.
It is a teacher with time.**

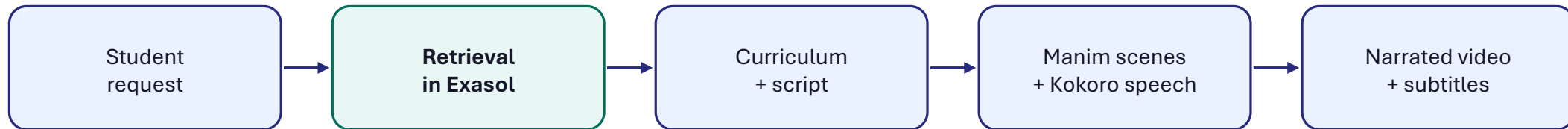
Education is where the need is most concrete and the impact most direct.
A supplement, not a replacement — an add-on that reaches the remotest
student and acts as a personalised tutor.

**Students get a second explanation on demand; teachers
get their scarcest resource back: time.**



Our solution

ScholarMotion turns a request written the way a student actually speaks into a narrated, animated lesson grounded in the NCERT textbook.



- No slide deck — a real animation with a real voice explaining it.
- Fully self-hosted speech, no per-minute API cost.

A request in. A grounded lesson out.

What makes it a tutor, not a video



A student watches, and at 1:24 they are lost. In a normal video that is the end of the road — you rewatch the same words that already failed you.

“from 1:24 to 2:30, explain this more slowly, with a simpler example.”

Scene	Span	Before	After
1	0:00–0:38	render 4	render 4 untouched
2	0:38–1:16	render 4	render 4 untouched
3	1:16–2:22	render 5	render 10
4	2:22–3:15	render 5	render 10

A tutor does not restart the lecture. A tutor re-explains the bit you did not get.

Where Exasol sits

Exasol Personal is our data platform for the knowledge corpus — the data the product is actually about.

What lives there

- Every NCERT chunk, with metadata and concept tags.
- Embeddings unrolled one row per dimension — 3,456 rows for our corpus.
- Long and narrow: the shape a columnar engine scans best.

What runs there

- Entire retrieval as SQL.
- Vector similarity as a columnar JOIN + SUM dot product.
- Blended with lexical overlap and concept-tag matching.
- Ranked in-database — only the top 5 rows return.

Scanning a whole corpus to score it is columnar analytics, not row lookups. And it is on the critical path — no lesson is planned before Exasol answers.

Multilingual by design



Every model sits behind a provider interface. The speech engine is one class with one method.

82M

Kokoro-82M runs locally:
laptop CPU, no API key,
no per-minute cost

.env

TTS_PROVIDER and
TTS_VOICE swap voices
by config

1× visuals

same animation can
serve Tamil, Hindi or
Bengali audio and
subtitles

- Narration is generated separately from the animation.
- Swapping to a multilingual voice is a config change, not a rewrite.

Plug and play. Finding a replacement multilingual model is all it takes — a child should not have to learn physics in their second language first.

Design choices that made this possible

01

Small immutable artifacts, not one big file.

Every scene has its own spec, code, audio and render, versioned independently. This is why sixty seconds can be regenerated without touching the rest.

02

Provider interfaces everywhere.

LLM, speech, embeddings. Swapping any one of them is configuration.

03

Exasol for the corpus, transactional store for state.

Scanning a corpus to score it is analytics; tracking scene versions is row-by-row bookkeeping. Different jobs, different engines.

04

Degrade, never crash.

No LaTeX? Emit Unicode. UDF unavailable? Fall back to columnar SQL. Exasol unreachable? Log it loudly and keep serving.

What we learned

Getting Exasol running taught us things a tutorial would not.

Reserved words

SECTION, TEXT and VALUE are reserved words — we learned this from a failing CREATE TABLE.

Pure-SQL fallback

Local Exasol Personal ships without the Python script container, so we built a pure-SQL columnar scoring path — and that fallback serves every query in this demo.

Every one of those we found by running against a real database, not by reading about one.

We thoroughly enjoyed this process, and we hope to learn more from the Exasol team in the future.

Summary



The need

1,10,971 single-teacher schools. 40 lakh children. 10 lakh vacant posts.

The product

A request in, a narrated animated lesson out — and a student can ask for any sixty seconds to be re-explained.

The platform

Exasol Personal holds the corpus and runs the whole retrieval as in-database SQL.

The reach

Self-hosted speech, provider-swappable models, multilingual by configuration.

Benefits everyone — students and teachers alike.

References and data sources

1 UDISE+ 2023–24, Ministry of Education, Government of India — single-teacher schools and enrolment.

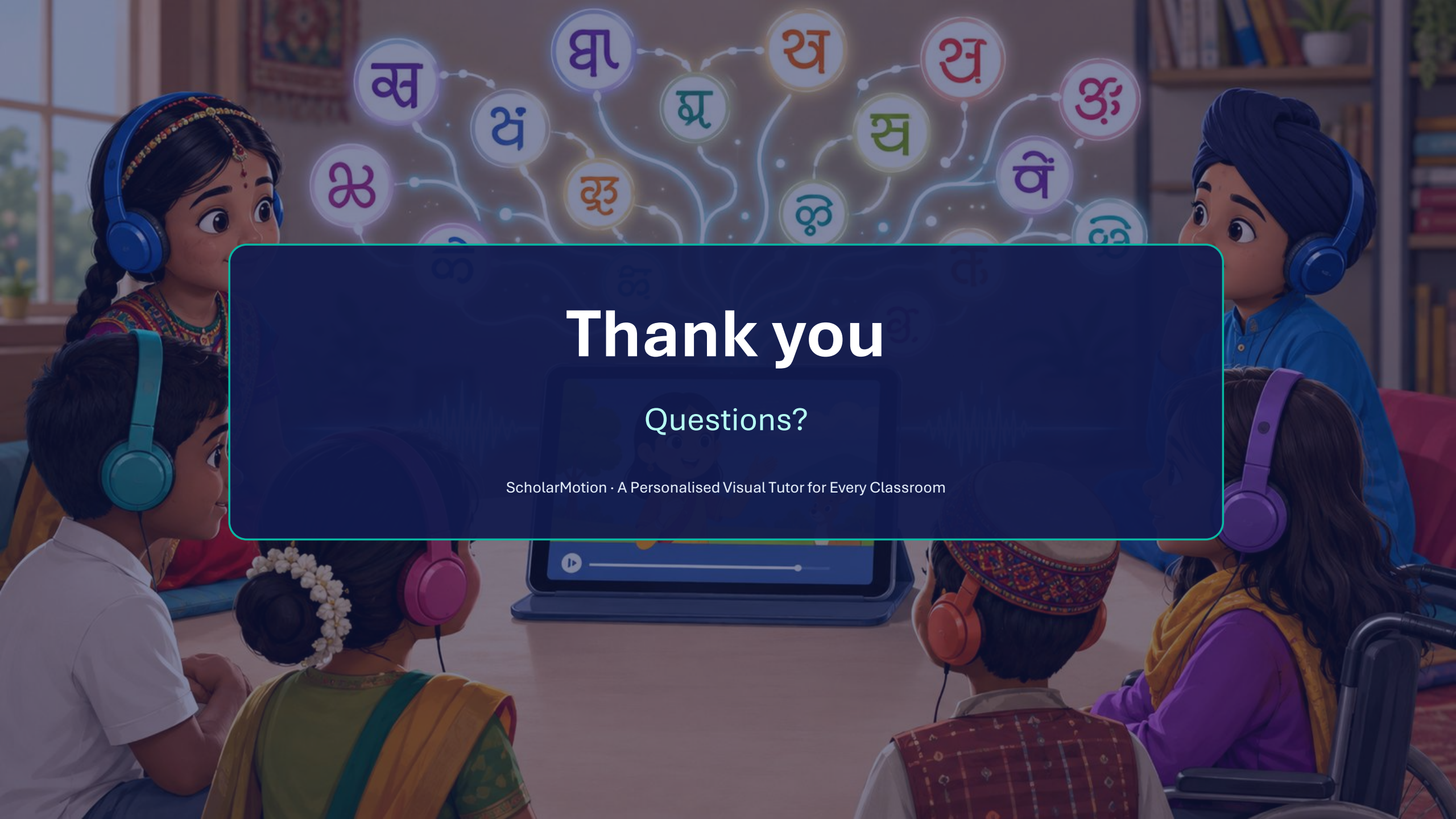
2 Teacher vacancy statistics, 2024–25 — vacancy rate and vacant sanctioned posts.

3 TRAI / DoT Telecom Statistics India 2025 — rural internet subscribers and density.

4 Exasol Personal — github.com/exasol/exasol-personal

5 Kokoro-82M (Apache-2.0) — huggingface.co/hexgrad/Kokoro-82M

6 Manim Community Edition — manim.community



Thank you

Questions?

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